A

G27D-K30H-Q58M
G27T-K30H-Q58M
G27T-K30H
G27D-K30H
G27T
G27D
G25E-K30H
K30H
G25E
A59P
M132D
M132E
K30H-Q58M
K60R

0
10
20
30
40

Deviation from WT (SD)

B

Pearson r = -0.151
n = 10

1.0
0.8
0.6
0.4
0.2
0.0

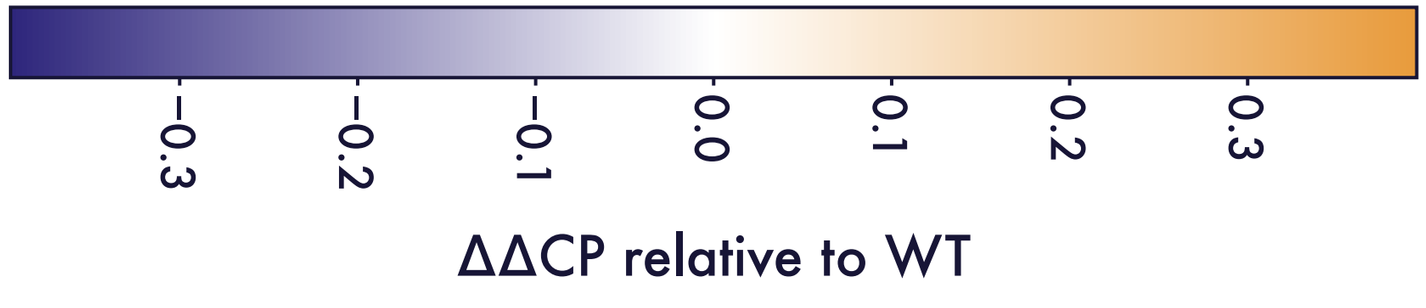
Activity retention (relative to parent)

0
10
20
30
40

Protein–target-C deviation (SD)

Internal CP (long-range)
Protein–target-C CP
Pocket CP
±3 SD band

	W130	F129	C105	C102	P101	W99	V71	H70	R69	G68	Y67	F66	L62	N61	K60	A59	Q58	N57	T31	K30	H29	R28
G25E	-0.0535	-0.0815	-0.071	-0.1485	-0.051	-0.1185	-0.0675	-0.1335	-0.0645	-0.068						0.1265		-0.118	-0.1355		0.165	-0.2
G25E-K30H															-0.078					0.18		
G27T		0.0685		0.0615		0.0615		0.0565							-0.098	-0.0635			0.0645	0.15		0.08
G27D															-0.088					0.15	0.085	0.05
G27T-K30H			0.059			0.0515	0.0525	0.0965		-0.098	-0.0595		-0.086	-0.09	-0.228		-0.0955		0.0545	0.32	0.235	0.12
G27D-K30H								0.0665		-0.088			-0.076	-0.08	-0.208		-0.0655			0.32	0.225	0.11
G27T-K30H-Q58M			0.069				0.0625	0.0965		-0.118	-0.0695	-0.0575	-0.086	-0.1	-0.228		-0.1055		0.0645	0.33	0.325	0.12
G27D-K30H-Q58M	-0.0735		0.059				0.0525	0.0765		-0.128	-0.0695	-0.0575	-0.076	-0.1	-0.228		-0.1055	-0.058		0.32	0.335	0.12
K30H								0.0665												0.16	-0.095	0.1
K30H-Q58M				0.0515				0.0665												0.05	-0.125	0.1
Q58M						0.0515																
R28A				0.0815				0.0865										0.052	0.0745		-0.125	0.14
R28S				0.0615				0.0865										0.052	0.0645	0.07	-0.105	0.12
T31V						0.0515													0.0545	0.14		
T31I							0.0725	0.0765					-0.076	-0.08	-0.198		-0.0555		0.0645	0.29	0.225	0.11
T31L										-0.088			-0.086	-0.1	-0.248		-0.1055		0.0745	0.38	0.395	0.14
Y32F																						
A59P								-0.0735	-0.0545									-0.078	-0.0855		0.055	0.05
K60R																						
M132D		-0.0815		-0.0885		-0.0885		-0.0735										-0.058	-0.0755		0.095	
M132E		-0.0915		-0.0985		-0.0985		-0.0835										-0.068	-0.0855		0.115	



A

B

WT 2.599 Å

A59P
G25E
G25E-K30H
G27D
G27D-K30H
G27D-K30H-Q58M
G27T
G27T-K30H
G27T-K30H-Q58M
K30H
K30H-Q58M
K60R
M132D
M132E
Q58M
R28A
R28S
T31I
T31L
T31V
Y32F
E72A
G27D-M130G
N57A
N57W
T31A
T31D
WT–GCG
WT–GCA

Distance unavailable

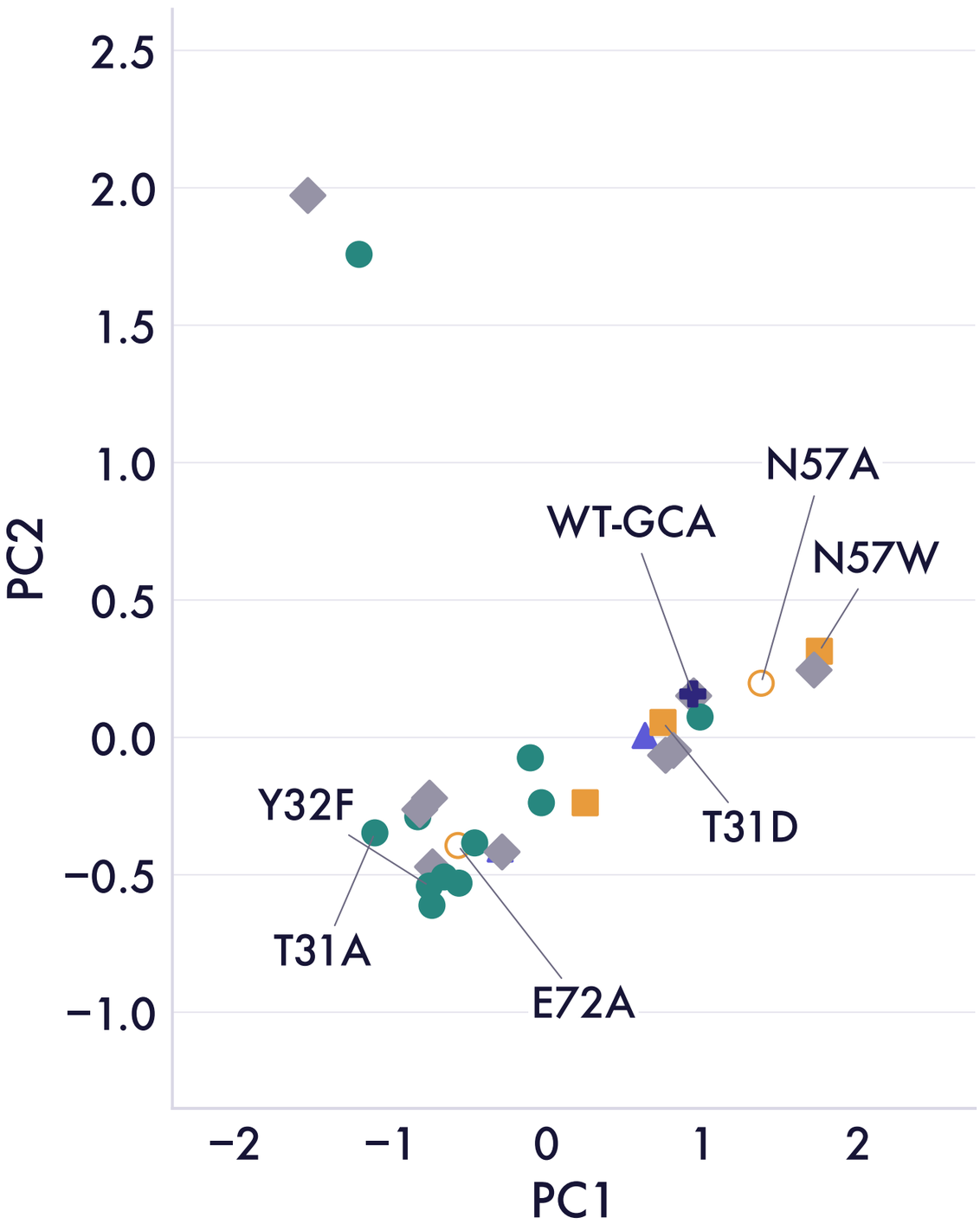
0
2
4
6
Target C4N4–E72 distance (Å)

0
20
40
60
Deviation from WT (SD)

Active
Inactive
Partially inactive
Not tested
Designed inactive control (not measured)
Wild-type reference

Internal CP (long-range)
Pocket CP
Protein–target-C CP
±3 SD band

A



Active

Inactive

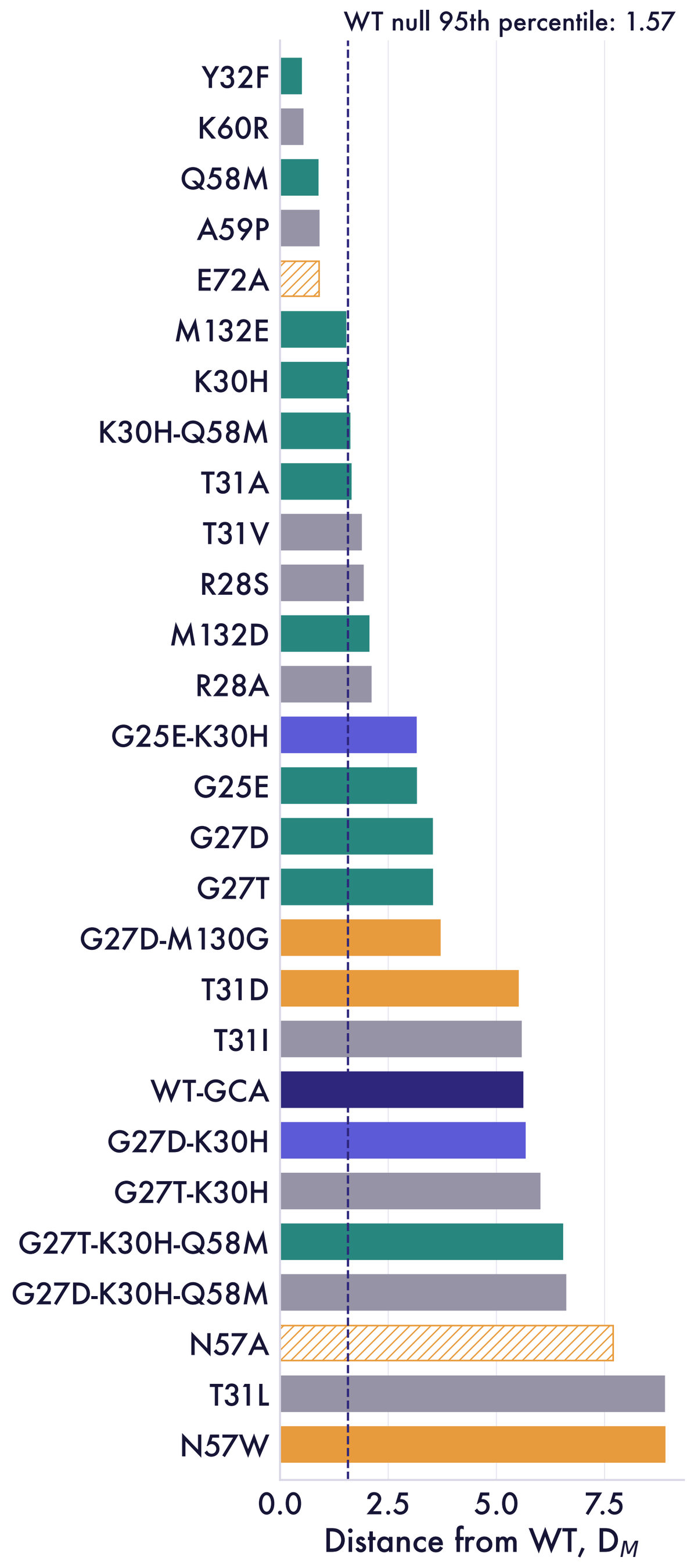
Partially inactive

Not tested

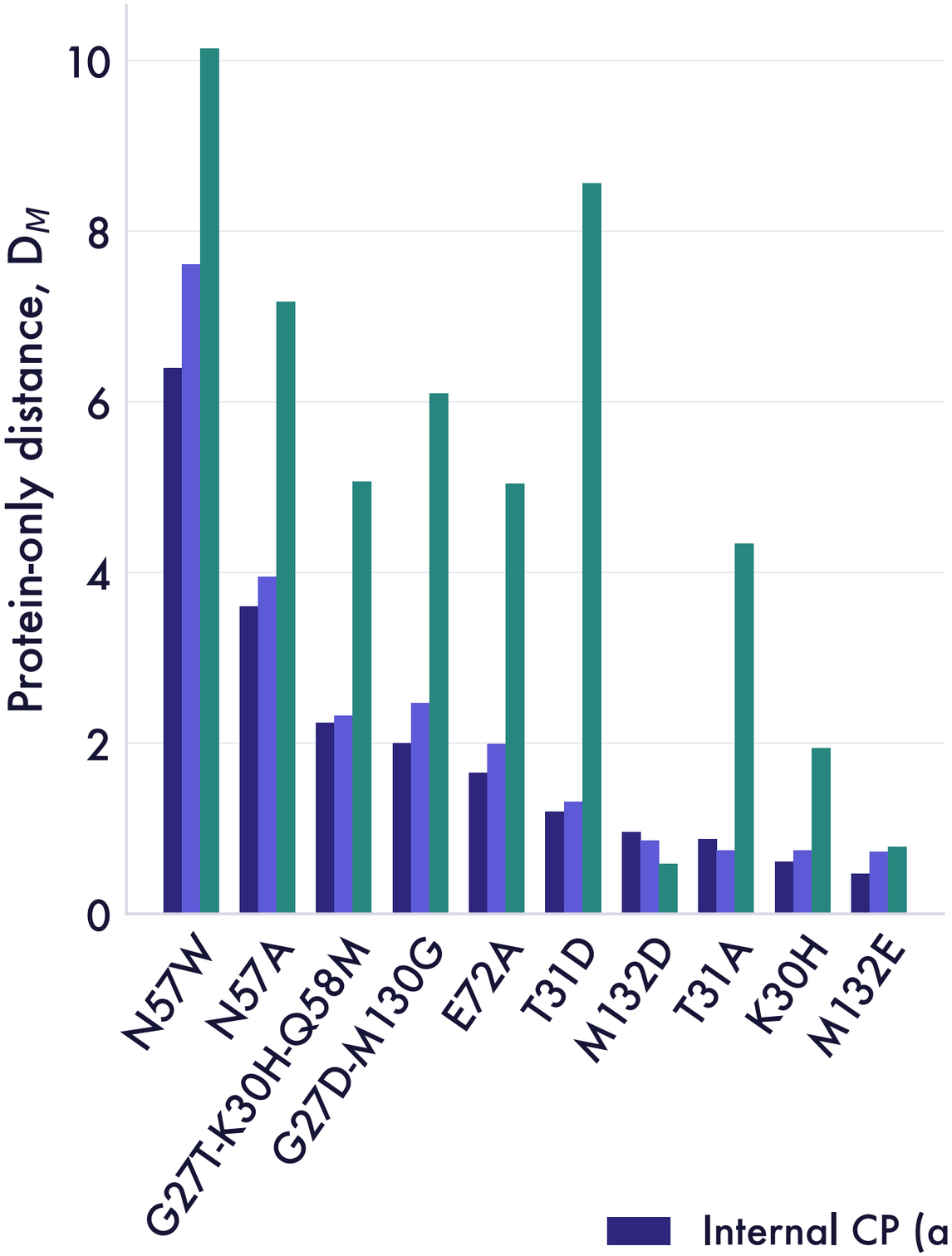
Designed inactive control (not measured)

Wild-type reference

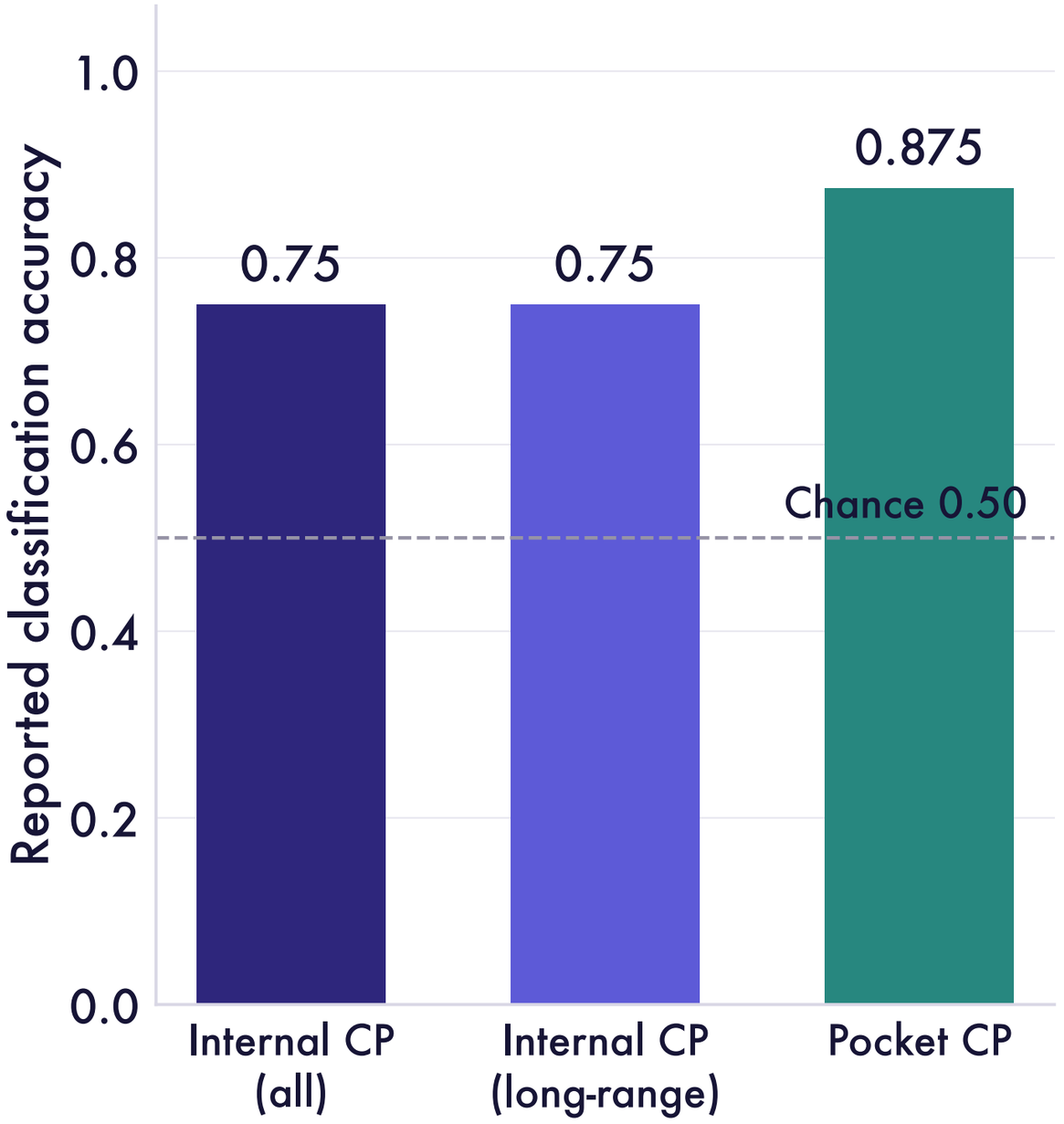
B



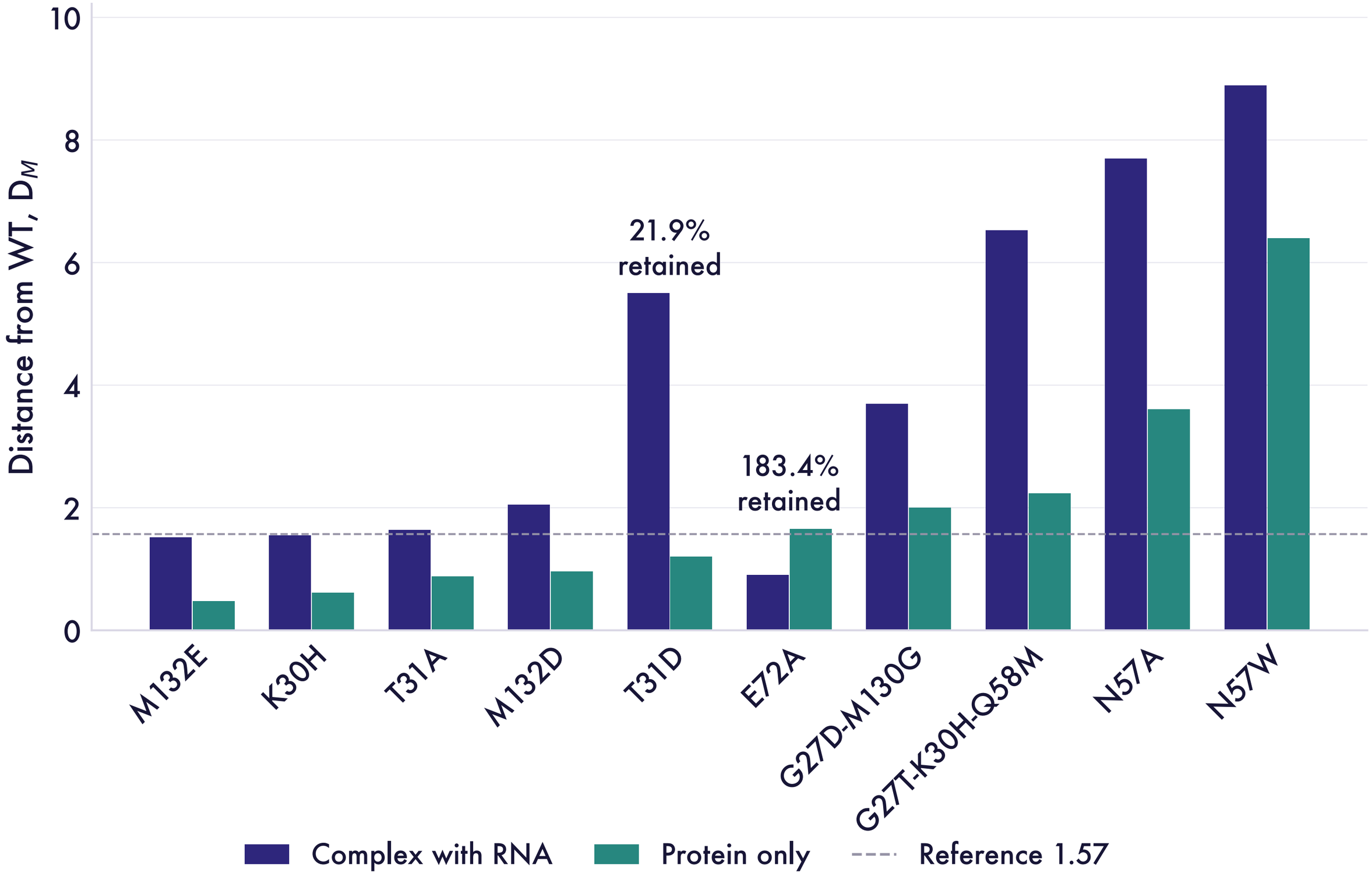
A

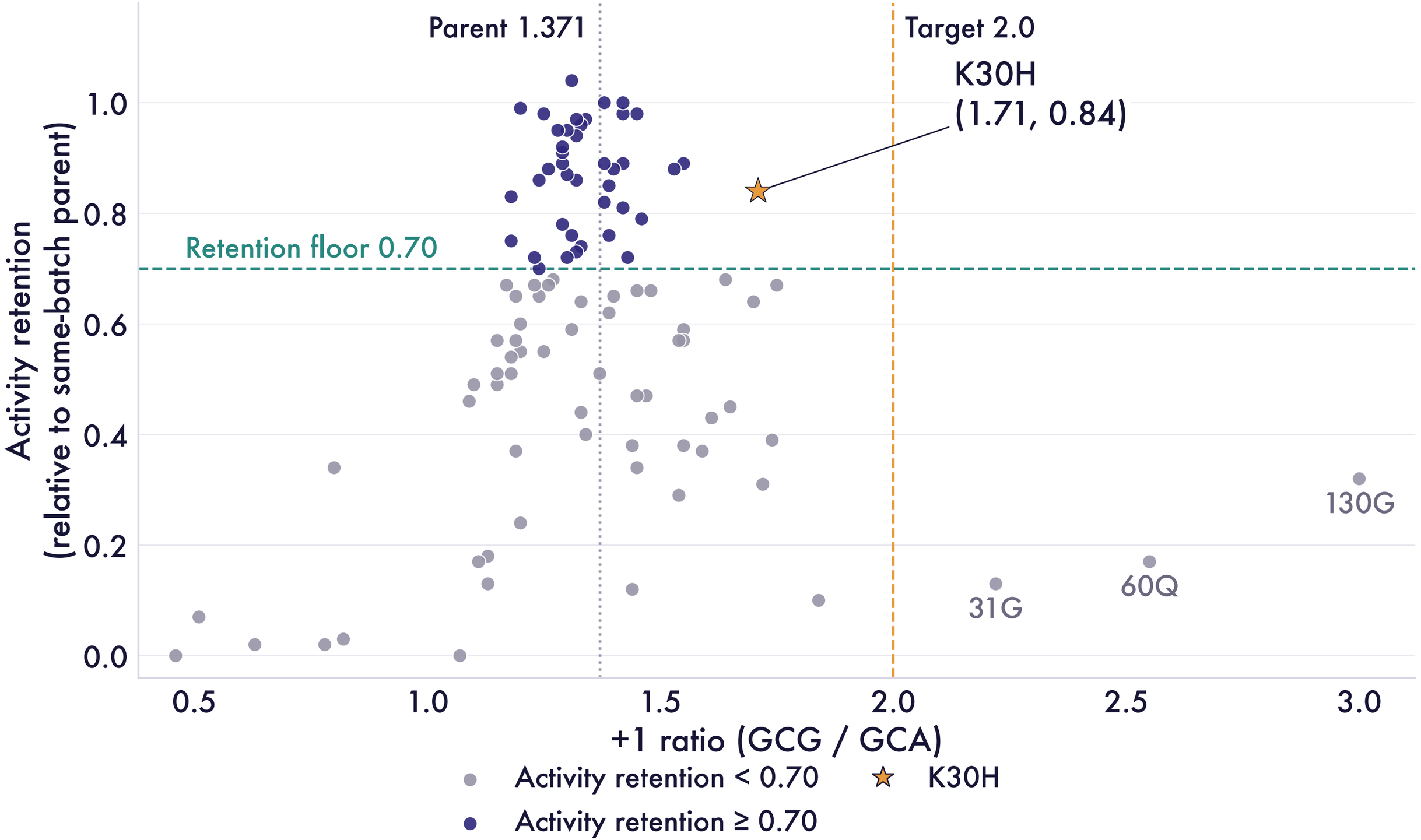

Protein-only distance, D_M
10
8
6
4
2
0
N57W
N57A
G27T-K30H-Q58M
G27D-M130G
E72A
T31D
M132D
T31A
K30H
M132E

B


Reported classification accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.75
0.75
0.875
Chance 0.50
Internal CP (all)
Internal CP (long-range)
Pocket CP


Internal CP (all)
Internal CP (long-range)
Pocket CP


Distance from WT, D_M
10
8
6
4
2
0
21.9%
retained
183.4%
retained
M132E
K30H
T31A
M132D
T31D
E72A
G27D-M130G
G27T-K30H-Q58M
N57A
N57W
Complex with RNA
Protein only
Reference 1.57


Parent 1.371
Target 2.0
K30H
(1.71, 0.84)
Retention floor 0.70
130G
60Q
31G
Activity retention
(relative to same-batch parent)
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
+1 ratio (GCG / GCA)
Activity retention < 0.70
K30H
Activity retention ≥ 0.70

A

Median
2.53

Median
1.20

Targeting ratio

On-target C388 editing (%)

10–20% on-target

35–60% on-target

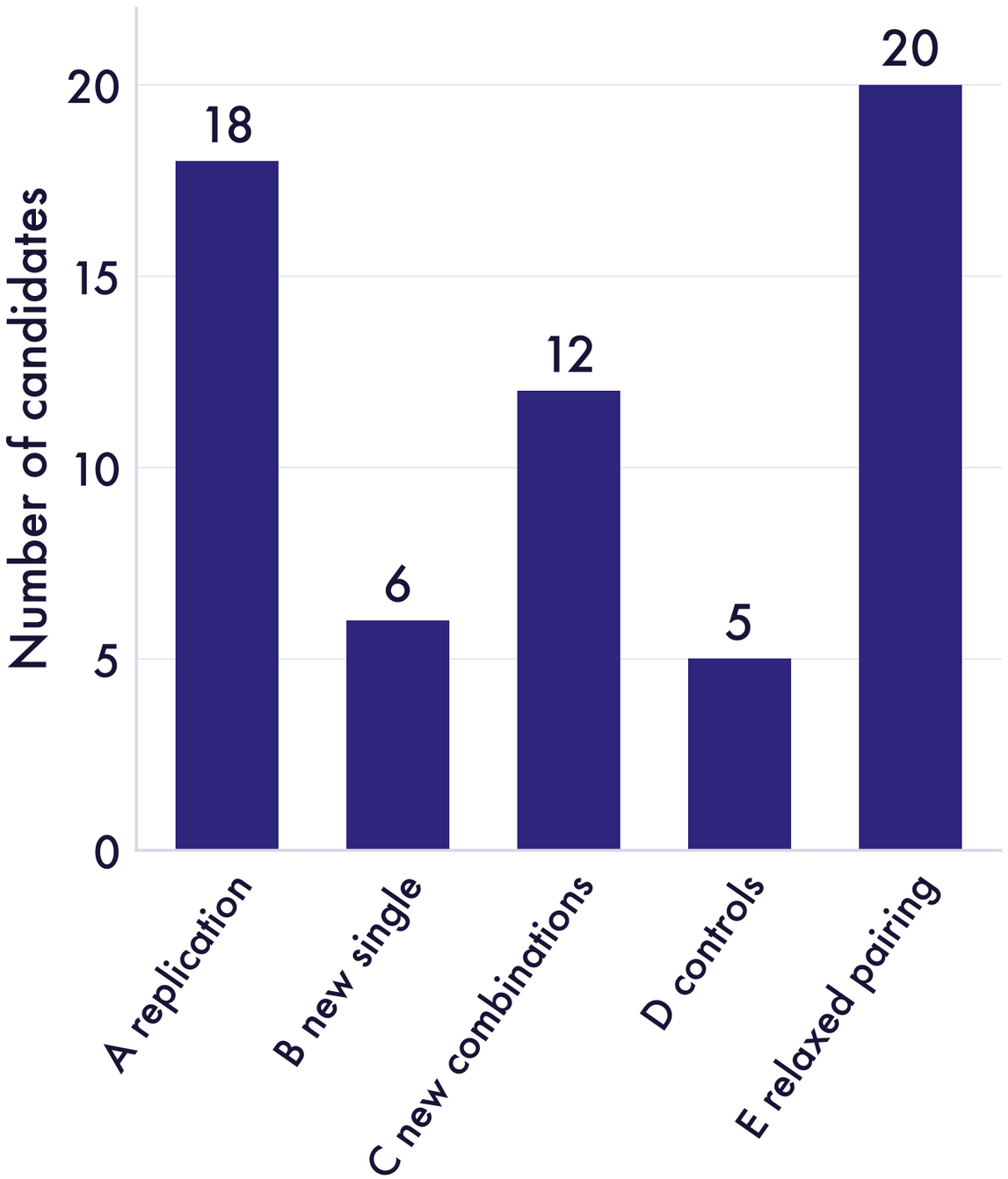
B

Pearson r = -0.390
Spearman ρ = -0.557
n = 41

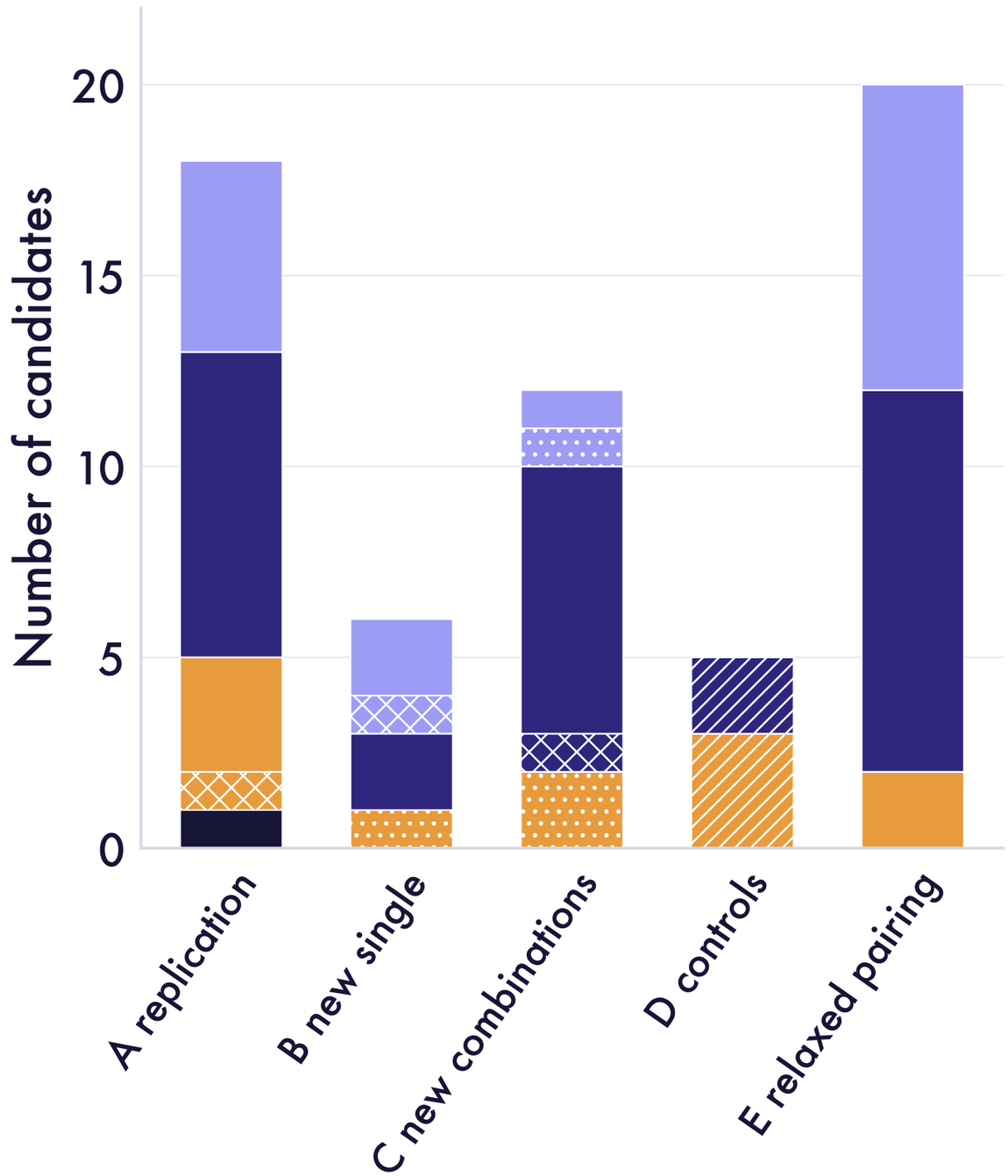
Summed non-GCG editing (%)

Linear fit (panel B)

A

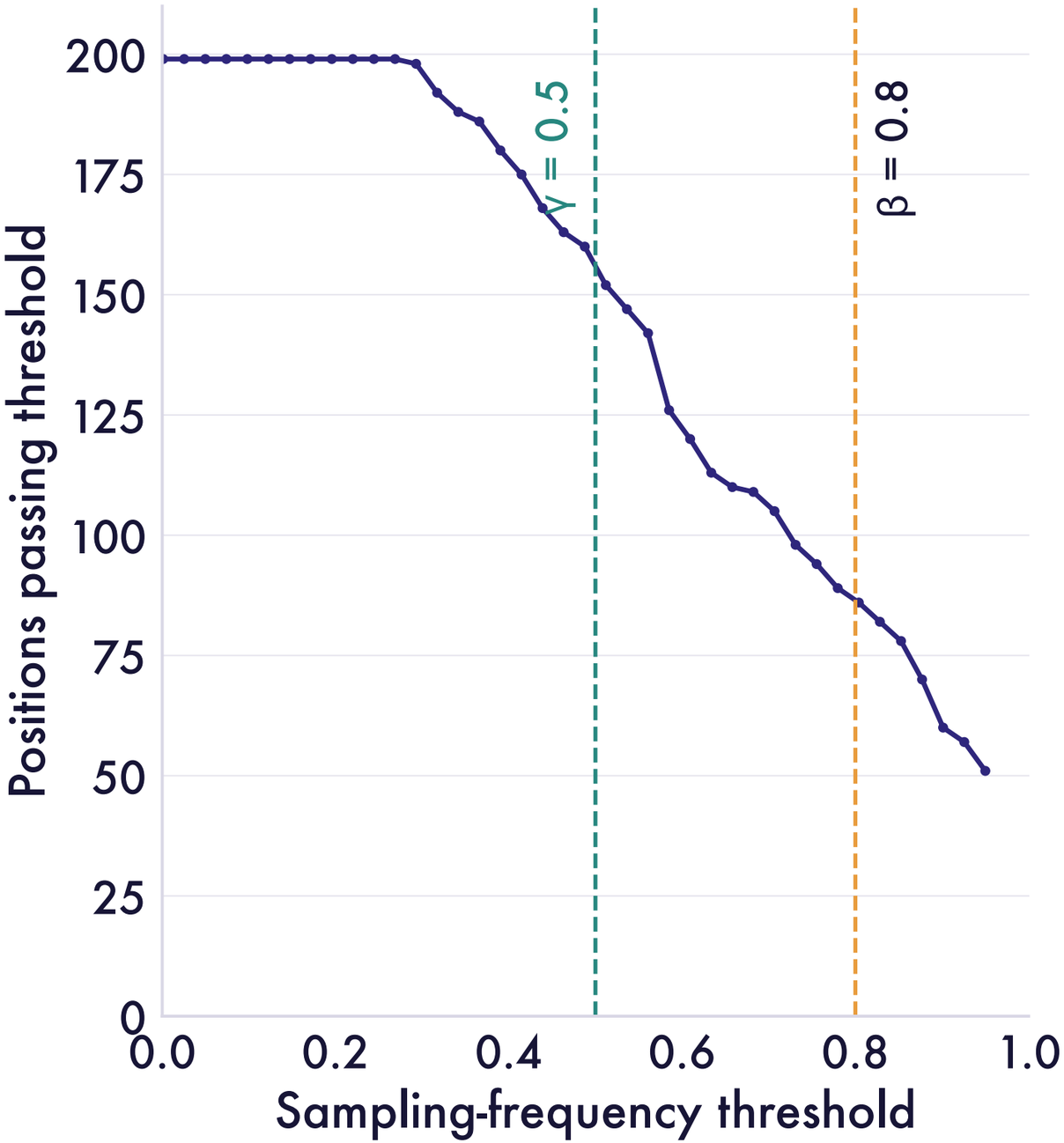


B

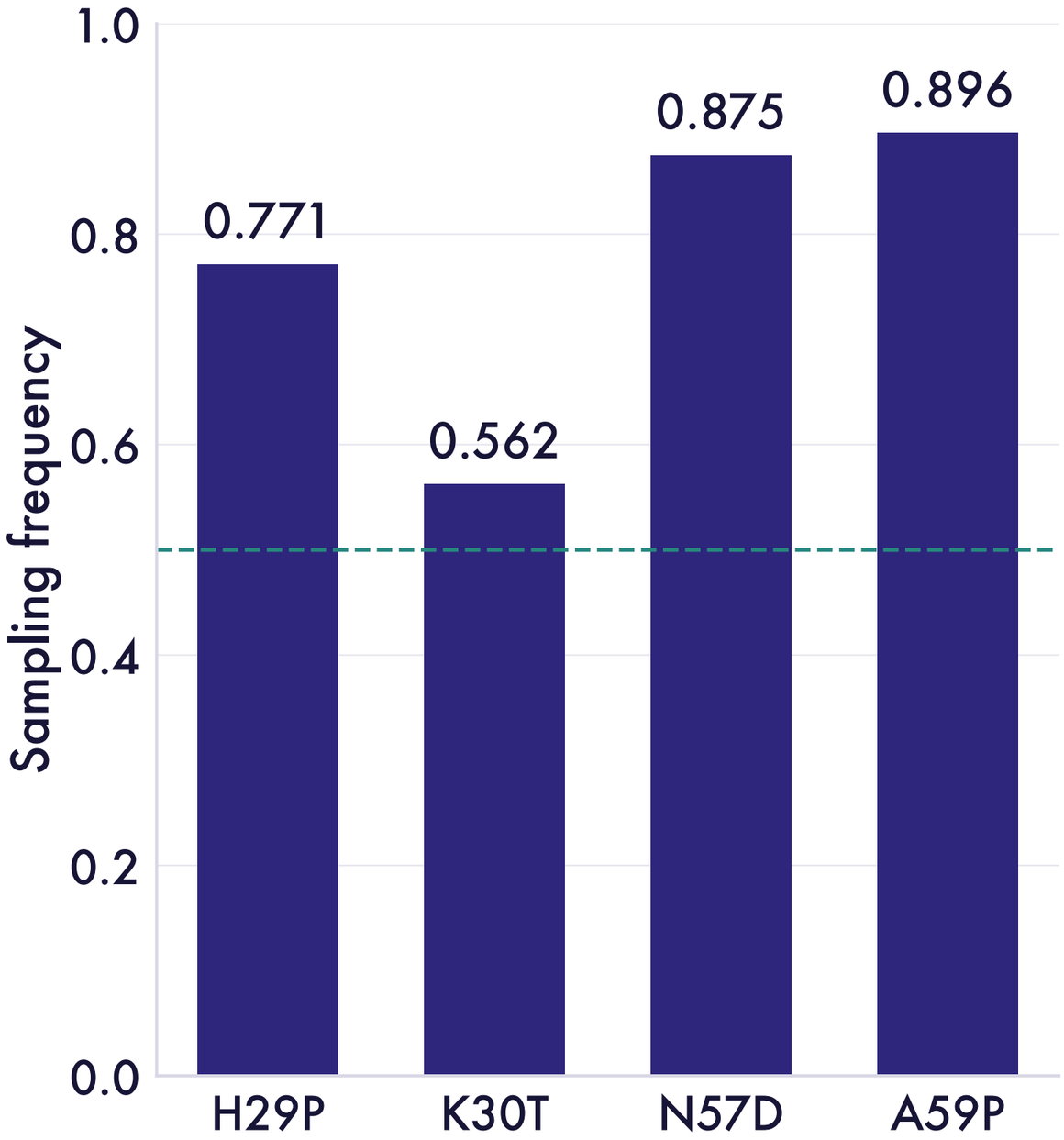


Required
P0 (top pick)
P0 (gold standard)
P0 (control)
P0
P1 (control)
P1 (mechanism)
P1
P2 (mechanism)
P2 (reference)
P2

A



B



Coil threshold γ = 0.5

Structured-region threshold β = 0.8